

Salifort Motors

Employee Attrition Analysis

ISSUE / PROBLEM

Given the high employee turnover rate at Salifort Motors, the HR department is looking for initiatives to improve satisfaction by identifying the core factors that lead to employee attrition.

They have the following question: **what's likely to make an employee leave the company?**

RESPONSE

Predicting employee attrition is a supervised binary classification task. The methodology included:

- **Baseline Model:** Logistic Regression established the initial prediction and impact of predictors.
- **Model Improvement:** Random Forest and XGBoost models were compared using validation F1 scores to enhance predictive power.
- **Final Selection:** XGBoost was chosen for its marginally superior performance.

IMPACT

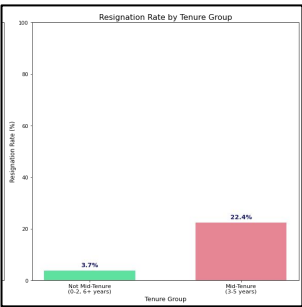
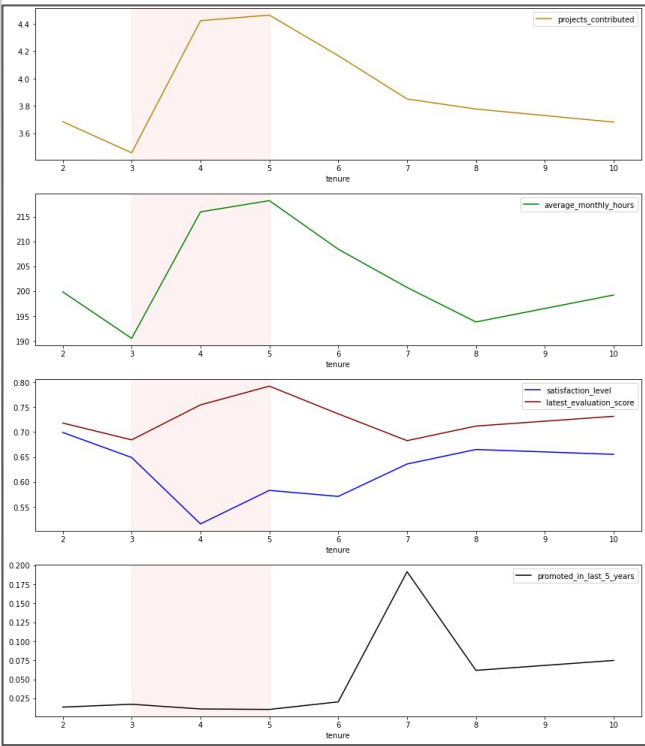
Exploratory Data Analysis (EDA) revealed several critical patterns in employee turnover.

Insights from the EDA were confirmed by the Logistic Regression and Machine Learning Models, which also identified the top predictors contributing to employee attrition. This provides management with a foundation for developing data-driven strategies.

KEY INSIGHTS

The story of Salifort Motors is one of unbalanced excellence. To increase retention, the focus shouldn't just be on the "bad" employees leaving, but on calibrating the workload for the "best" ones before they resign

- **Cap Project Assignments:** Limit projects to 5 max per employee; turnover spikes at 6.
- **Monitor Work Hours:** Alert managers when employees consistently exceed 200–210 hours/month.
- **Focus on 3–5 Year Tenure:** Proactively support and recognize employees in this high-risk bracket. Link exceptional performance to career advancement or financial incentives. Immediately conduct focus groups with this cohort to gather insights and develop job satisfaction strategies.
- **Investigate Low-Engagement Groups:** Address low satisfaction among employees with only 2 projects; they may need more challenging work or better team integration.



- Mid-tenure employees (those with 3-5 years of tenure) have a significantly higher resignation rate compared to other employees

- The company is successfully identifying and utilizing its "Champions" (the high-project, high-evaluation group), but is failing to protect them from burnout and rewarding them adequately.
- The XGBoost predicts that the company's top performers (likely to be in the mid-tenure)—those taking on the most projects, working the most hours, and receiving the highest marks—are the ones most likely to leave

